

PROJECT AND TEAM INFORMATION

Project Title

Internet-Free Messenger: A Bluetooth-Based Chat Application

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Popular chat applications such as WhatsApp depend entirely on an active internet connection. In locations with weak network coverage, no data connectivity, or during emergencies, this dependency makes communication unreliable or completely unavailable.

To address this, our project builds a lightweight, WhatsApp-style messaging application that communicates over **Bluetooth instead of the internet**. Two or more nearby devices can pair directly and exchange text messages without relying on mobile data, Wi-Fi, or any external network.

Such a system is especially valuable in **college campuses, hostels, remote regions, and disaster or emergency scenarios** where connectivity cannot be guaranteed. Building it also gives us a hands-on opportunity to apply **C/C++, Object-Oriented Programming, and core Data Structures and Algorithms** – including queues, linked lists, and searching – to a real communication problem.

State of the Art / Current solution

Today's messaging platforms – WhatsApp, Telegram, Messenger, and similar apps – are all built around continuous internet or mobile-data access. When that access drops, whether due to poor signal, data exhaustion, or an outage, these applications stop functioning altogether.

Bluetooth and Wi-Fi Direct already exist as short-range, internet-free communication technologies, but they are rarely packaged into a simple, chat-style interface the way WhatsApp is. This leaves a clear gap: an approachable, offline messaging tool that lets nearby users talk to each other purely through Bluetooth, without any supporting infrastructure.

Project Goals and Milestones

General Goal: Build a compact, WhatsApp-inspired chat application that lets two or more nearby devices exchange text messages over Bluetooth alone, with no internet dependency.

Initial Milestones: Research how Bluetooth pairing and data transfer work; sketch the application's interface and control flow; implement device discovery and a working connection; apply C/C++ and DSA concepts to manage messages internally.

Further Milestones: Layer in chat history and persistent message storage; extend the system toward multiple nearby devices; evaluate range, speed, and connection reliability; resolve bugs and polish the user experience; wrap up with full testing, documentation, and a project demonstration.

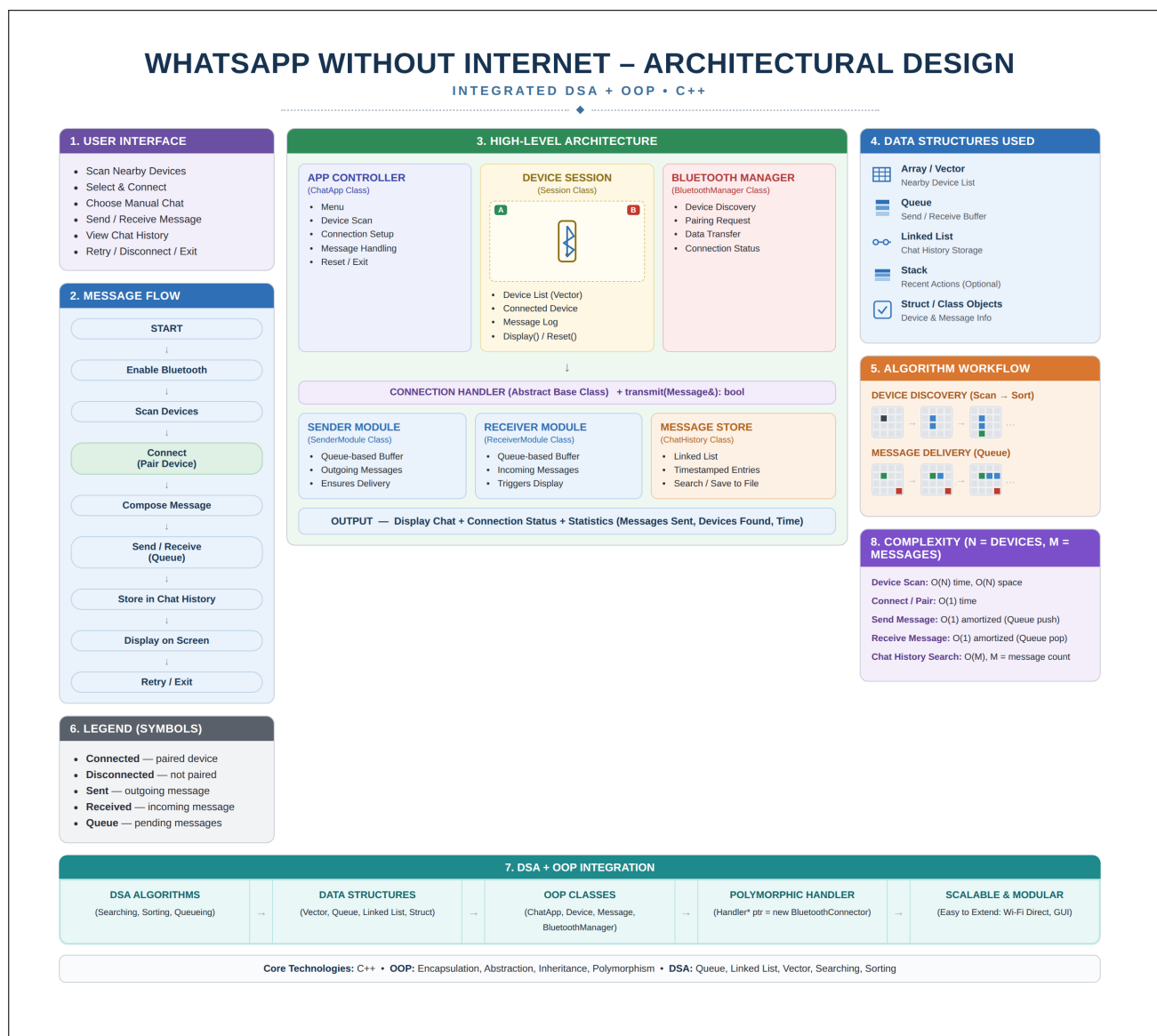
Project Approach

Our approach is to build an offline chat system that feels like a stripped-down WhatsApp but runs entirely on **Bluetooth instead of the internet**. The interface will let a user scan for nearby devices, initiate a connection, and start exchanging messages once paired.

Internally, the application is written in **C/C++** and leans on **Data Structures and Algorithms** to manage users, in-flight messages, and chat history – using **queues, linked lists, searching, and sorting** wherever they fit naturally. **Object-Oriented Programming** – classes, objects, inheritance, and polymorphism – keeps the codebase organized as the system grows.

On the connectivity side, we rely on **Bluetooth-enabled hardware** together with the appropriate Bluetooth communication libraries or APIs, with an initial prototype targeting **Windows and Android**. Testing will involve multiple physical devices to validate pairing, message delivery, local storage, and overall reliability.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

By the end of this project we will have a working **Bluetooth-powered offline messaging application**. Users will be able to find nearby devices, pair with them over Bluetooth, and exchange text messages through a simple chat screen, with messages saved locally as chat history. Deliverables include a functioning prototype, the complete source code, supporting project documentation, a record of testing results, and a final presentation of the system.

Assumptions

- Users have Bluetooth-enabled devices.
- Devices are within the Bluetooth communication range.
- Users have Bluetooth turned on and allow device connections.
- The system will mainly support text messages in the initial version.
- No internet, mobile data, or Wi-Fi connection is required.
- Devices have enough storage to save basic chat history.
- The project will be tested with a limited number of nearby users.

References

- Bluetooth Special Interest Group (Bluetooth SIG) – Bluetooth Technology Overview.
- Herbert Schildt – *C++: The Complete Reference*, McGraw-Hill Education. Useful for C++ and OOP concepts.
- Mark Allen Weiss – *Data Structures and Algorithm Analysis in C++*, Pearson. Useful for DSA concepts such as linked lists, queues, searching, and sorting.
- Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed – *Fundamentals of Data Structures in C*, Universities Press.
- Android Developers – Bluetooth Connectivity Documentation.